

REVIEW ARTICLE (META-ANALYSIS)

Clinical Benefits of Joint Mobilization on Ankle Sprains: A Systematic Review and Meta-Analysis



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Abstract

Objective: To assess the clinical benefits of joint mobilization for ankle sprains.

Data Sources: MEDLINE, MEDLINE In-Process, Embase, AMED, PsycINFO, CINAHL, Cochrane Library, PEDro, Scopus, SPORTDiscus, and Dissertations and Theses were searched from inception to June 2017.

Study Selection: Studies investigating humans with grade I or II lateral or medial sprains of the ankle in any pathologic state from acute to chronic, who had been treated with joint mobilization were considered for inclusion. Any conservative intervention was considered as a comparator. Commonly reported clinical outcomes were considered such as ankle range of movement, pain, and function. After screening of 1530 abstracts, 56 studies were selected for full-text screening, and 23 were eligible for inclusion. Eleven studies on chronic sprains reported sufficient data for meta-analysis.

Data Extraction: Data were extracted using the participants, interventions, comparison, outcomes, and study design approach. Clinically relevant outcomes (dorsiflexion range, proprioception, balance, function, pain threshold, pain intensity) were assessed at immediate, short-term, and long-term follow-up points.

Data Synthesis: Methodological quality was assessed independently by 2 reviewers, and most studies were found to be of moderate quality, with no studies rated as poor. Meta-analysis revealed significant immediate benefits of joint mobilization compared with comparators on improving posteromedial dynamic balance ($P=.0004$), but not for improving dorsiflexion range ($P=.16$), static balance ($P=.96$), or pain intensity ($P=.45$). Joint mobilization was beneficial in the short-term for improving weight-bearing dorsiflexion range ($P=.003$) compared with a control.

Conclusions: Joint mobilization appears to be beneficial for improving dynamic balance immediately after application, and dorsiflexion range in the short-term. Long-term benefits have not been adequately investigated.

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Ankle sprains are a common injury in sports and the general community, and may lead to chronic pain, functional limitations, and physical disability.^{1,2} Epidemiologic studies conducted in various countries highlight the high incidence of ankle sprains during sports training and competition, with rates reported as 7 per

1000 in Denmark, 6.09 per 1000 in the United Kingdom, and 2.15 per 1000 in the United States in person-years.³⁻⁵ Plantarflexion inversion sprain or lateral ankle sprain is the most common type of ankle sprain.⁶ It typically results in an injury of the inferior tibiofibular ligament, anterior tibiofibular ligament, or the bifurcate ligament.⁷ Eversion injuries often result in damage to the deltoid and spring ligaments of the medial aspect of the ankle.⁷

According to the clinical practice guidelines linked to the *International Classification of Functioning, Disability and Health* from the Orthopaedic Section of the American Physical Therapy Association, manual therapy is recommended for both the acute and progressive loading phases of rehabilitation.⁸ Management of

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ankle sprains commonly involves mobilization procedures applied to the joint, such as nonthrust joint mobilization, high-velocity thrust manipulation, and mobilization with movement (MWM).

The mechanisms by which these techniques are purported to work are biomechanical (such as stretching/tearing tissue, inducing cavitation within the joint, reducing muscle hypertonicity/stiffness) and neurophysiological, potentially including spinal cord and supraspinally mediated mechanisms.^{9,10}

Several studies¹¹⁻²¹ have investigated the effects of manual therapy on ankle sprains using a variety of outcome measures including pain, range of motion (ROM), and function from the acute to chronic stages of recovery, with different results reported. Several systematic reviews²²⁻²⁶ have attempted to collate this evidence but have been limited by their narrow focus on lateral ankle sprains and restricted outcome measures. Previous systematic reviews have all included some studies that involved other interventions such as “rest, ice, compression, and elevation” (RICE) and home exercise programs as an adjunct to mobilization. Therefore, they have not actually assessed mobilization as the sole intervention. Moreover, the clinical benefits of joint mobilization have not yet been evaluated through meta-analysis, despite it being a common intervention used in the rehabilitation of a number of ankle conditions and despite the growing body of empirical literature.

The present systematic review aims to address these limitations by synthesizing and meta-analyzing the available evidence for ankle joint mobilization (including high-velocity thrust manipulation) in grade I or II ankle sprains of the medial or lateral ligaments in the acute/subacute/chronic stages of rehabilitation in any ambulant setting.

Methods

Registration

The protocol for this systematic review was registered with the International Prospective Register of Systematic Reviews on January 12, 2016 (CRD42016030194).

Search strategy

A search of electronic databases, including MEDLINE, MEDLINE In-Process, Embase, AMED, PsycINFO, CINAHL, Cochrane Library, PEDro, Scopus, SPORTDiscus, and Dissertations and Theses was conducted from inception to June 2017. In addition to the database search, a hand search of the reference lists of identified studies was also carried out. A search strategy (supplemental appendix S1, available online only at <http://www.archives-pmr.org/>) was developed for the main search strings of ankle sprain and mobilization. Keywords used for “ankle sprain” included sprain, talocrural joint, ligament injuries, lateral

ligament, medial ligament, deltoid ligament, collateral ligament, anterior talofibular ligament, posterior talofibular ligament, sprain and strain, and ankle twist. Keywords used for “mobilization” included manual therapy, joint mobilization, manipulation, MWM, Maitland, Mulligan, and rehabilitation. These terms were used alone and in combinations during the search.

Identification and selection of studies

Full-text randomized controlled trials, crossover studies, cross-sectional studies, cohort studies, and case series published in peer-reviewed journals and dissertations were considered for the present review. Studies were not restricted by language, provided the title and abstract were in English. Studies not involving live human participants (eg, model-based, animal, and cadaveric investigations) were excluded. Conference proceedings, commentaries, research notes, editorials, and letters were also excluded. To be included, studies were required to meet the criteria that follow.

Participants

Participants were live humans (without any age limitation) with a grade I or II lateral or medial ligament sprain of the ankle at any stage of recovery (acute to chronic) in any ambulant setting who have been treated with joint mobilization. Studies involving grade III sprains, fractures (other than Weber type A), and syndesmotic injuries were excluded from this review.

Intervention

Studies reporting any type of joint mobilization techniques applied to the talocrural joint, subtalar joint, or inferior tibiofibular joint by a physiotherapist, medical practitioner, osteopath, chiropractor, or athletic trainer were eligible for inclusion in the review. Interventions other than therapist-performed joint mobilization were excluded from the review.

Comparators

Studies reporting any conservative intervention for comparison, such as exercise therapy, elevation and icing, supportive strapping, sham intervention, or no treatment, were eligible for inclusion. Control groups with healthy subjects were also eligible as a comparator. Studies that compared mobilization techniques to surgical interventions were excluded.

Outcome measures

All commonly reported clinical impairments (pain, swelling, balance, proprioception, strength, stability, gait), activity restriction and self-reported confidence, community participation, quality of life, reinjury rate, function, and return to sport were considered for the review. The primary outcomes of interest were ankle ROM, pain, quality of life, and function.

Timing of the measurement of the outcomes was categorized as “immediate,” measured immediately after the intervention²⁷; “short-term,” measured up to 3 months after the intervention²⁸; and “long-term,” measured at 3 or more months²² after the intervention.

Identified studies were exported to reference management software (EndNote X7.3.1²⁹), and duplicate records were manually removed. Study titles and abstracts were initially screened by 2 independent reviewers, followed by screening of full-text articles, to determine the eligibility of the identified studies. Disagreement between the reviewers was resolved by consensus or involvement of a third reviewer. The level of agreement between reviewers was assessed using Cohen’s kappa.²⁹

List of abbreviations:

CI	confidence interval
DFROM	dorsiflexion range of motion
MCID	minimal clinically important difference
MD	mean difference
MWM	mobilization with movement
RICE	rest, ice, compression, and elevation
ROM	range of motion
SEBT	star excursion balance test
SMD	standardized mean difference

Table 1 Description of eligible studies

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Alanson et al, ³⁶ 2012	RCT	17 (10M) Grade 1/2 Chronic lateral ankle sprains	TCJ (anteroposterior) mobilization + TCJ traction 30s	Sham	Immediate	Non—weight-bearing DFROM, proprioception (joint position sense)	Non—weight-bearing DFROM, significantly improved across time ($P=.04$). Joint position sense significantly improved across time at target angle 10° plantarflexion ($P=.03$).
Beazell et al, ³⁷ 2012	RCT	43 Chronic ankle sprains	Distal TFJ manipulation + HVLA thrust 1 repetition	No intervention	Immediate, short-term (1, 2, and 3 wk)	Weight-bearing DFROM, static balance (single-limb stance), function (step down test, self-reported function, FAAM sports)	Weight-bearing DFROM not significant ($P=.82$). Single-limb stance not significant ($P=.42$). Function not significant—step down test ($P=.76$), self-reported function ($P=.61$), FAAM sports ($P=.83$)
Collins et al, ¹³ 2004	Randomized crossover	16 (8M) Grade 2 Subacute lateral ankle sprains	Weight-bearing MWM TCJ (posterior talar glide, posteroanterior tibial glide) 3 sets of 10 repetitions	Placebo No intervention	Immediate	Weight-bearing DFROM, pressure-pain threshold, thermal pain threshold	Weight-bearing DFROM significantly improved across time ($P=.013$) and no significant group difference (vs placebo, $P=.202$; vs control, $P=.208$). Pressure-pain threshold and thermal pain threshold not significant ($P<.05$).
Cruz-Díaz et al, ¹⁴ 2015	RCT	81 (47M) Chronic ankle sprains	Weight-bearing MWM TCJ (posterior talar glide, posteroanterior tibial glide) 2 sets of 10 repetitions, 2 sessions/wk for 3wk	Sham, no intervention	Immediate, short-term (3wk), long-term (6mo)	Weight-bearing DFROM, dynamic balance (SEBT)	Weight-bearing DFROM significantly improved— $P<.0001$ (at each time point). Dynamic balance significantly improved— $P<.0001$ (each direction of SEBT)
Gilbreath et al, ²¹ 2014	Prospective longitudinal	11 (5M) Chronic ankle sprains	Weight-bearing MWM TCJ (posterior talar glide, posteroanterior tibial glide) 2 sets of 4 repetitions 4min of MWM×3 sessions over 1wk	No control group	Short-term (after 24–48h)	Weight-bearing DFROM, dynamic balance (SEBT), function (FAAM)	Weight-bearing DFROM not significant ($P=.69$). Dynamic balance not significant (SEBT-anterior, $P=.99$; posteromedial, $P=.15$; posterolateral, $P=.24$). FAAM ADL not significant ($P=.19$). FAAM Sports significantly improved across time ($P=.01$).
Harkey et al, ³⁸ 2014	RCT	30 (14M) Chronic ankle sprains	Maitland mobilization TCJ (anteroposterior grade III) 3 sets of 60s	No intervention	Immediate	Non—weight-bearing DFROM, dynamic balance (SEBT)	Non—weight-bearing DFROM significantly improved ($P=.049$). Dynamic balance no improvement ($P>.05$).

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Table 1 (continued)

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Hoch and McKeon, ³⁹ 2011	Randomized crossover	20 (9M) Chronic ankle sprains	Maitland mobilization TCJ (anterior posterior grade III) 50±5 of 1-s oscillations ×2	No intervention	Immediate	Weight-bearing DFROM, static balance, dynamic balance (SEBT), talar stiffness	Weight-bearing DFROM significantly improved ($P=.01$). Static balance significantly improved. Time to boundary anteroposterior minima significantly improved ($P<.0001$). Dynamic balance—not significant— $P=.98$ (normalized reach distance). Talar stiffness not significant ($P=.08$).
Hoch et al, ⁴⁰ 2012	Prospective longitudinal	12 (6M) Chronic ankle sprains	Maitland mobilization TCJ (anteroposterior grade III) + TCJ traction 2 sets of 2min traction and 4 sets of 2min mobilization	No control group	Short-term (24–48h and 1-wk follow-up)	Weight-bearing DFROM, dynamic balance, function (FAAM)	Weight-bearing DFROM significantly improved across time ($P<.0001$). Dynamic balance significantly improved across time (SEBT anterior, $P<.0001$; posteromedial, $P=.003$; posterolateral, $P<.0001$). FAAM ADL and Sports significantly improved across time ($P=.001$).
Hoch et al, ⁴¹ 2014	Prospective longitudinal	12 (6M) Chronic ankle sprains	Maitland mobilization TCJ (anteroposterior grade III) + TCJ traction 2 sets of 2min traction and 4 sets of 2min	No control group	Short-term (24–48h, and 1-wk follow-up)	Static balance, talar stiffness	Static balance not significant. Time to boundary anteroposterior and time to boundary mediolateral not significant ($P>.05$). Talar stiffness not significant ($P>.05$).
Hopper et al, ⁴² 2009	Randomized controlled Within-subjects design	20 (8M) Chronic ankle sprains	Mulligan ankle taping Not explicitly stated	Injured taped, Injured untaped, Uninjured taped, Uninjured untaped	Immediate	Static balance, dynamic balance (wandering, overshoot, reaction time)	Static balance significantly improved in postural sway recovery across time ($P<.001$). Single-limb stance not significant ($P=.792$), Dynamic tracking balance not significant; wandering ($P=.559$), overshoot ($P=.547$), reaction time ($P=.142$).
Houstan et al, ⁴³ 2013	Prospective longitudinal	12 (6M) Chronic ankle sprains	Maitland mobilization TCJ (anteroposterior grade III) + TCJ traction 4min of traction and 8min of mobilization 6 sessions over 2wk.	No control group	Immediate, short-term (1-wk follow-up)	Function (FAAM Sports)	FAAM ADL some components significantly improved across time; walking on even ground ($P=.06$); going down stairs ($P=.07$); walking on uneven ground ($P=.03$); light to moderate work ($P=.06$); heavy work ($P=.03$); recreational activity ($P=.07$).

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Table 1 (continued)

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Joseph et al, ⁴⁴ 2010	RCT	40 (19M) Grade 1/2 Chronic lateral ankle sprains	Ankle axial elongation TCJ (superior inferior)-HVLA thrust 6 sessions over 3wk	Muscle energy technique	Short-term (1mo)	DFROM, plantarflexion ROM, static balance, pain quality and intensity, function (Functional Evaluation Scale)	FAAM Sports some components significantly improved across time; landing ($P=.03$); low impact activities ($P=.07$); cutting ($P=.02$). DFROM significantly improved across time ($P<.001$) and no significant group differences ($P=.713$). Plantarflexion ROM significantly improved across time ($P<.001$) and no significant group differences ($P=.300$). Single-limb stance—eyes closed significantly improved across time ($P<.001$) and no significant group differences ($P=.344$). Single-limb stance—eyes open significantly improved across time ($P<.001$) and no significant group differences ($P=.413$). McGill significantly improved across time ($P<.001$) and no significant group differences ($P=.077$). Functional Evaluation Scale significantly improved across time ($P<.001$) and no significant group differences ($P=.144$).
Kohne, et al, ⁴⁵ 2007	RCT	30 (21M) Grade 1/2 Chronic, recurrent lateral ankle sprains	Ankle axial elongation TCJ (superior inferior by a mortise separation) —(6 manipulations over 4wk)	Single manipulation treatment	Short-term (fifth week follow-up)	DFROM, proprioception (joint position sense), pressure-pain threshold, pain intensity	DFROM significantly improved: $P=.028$ (across time). Joint position sense at 5° plantarflexion error significantly improved: $P=.029$ (across time). Pressure-pain threshold (P value not reported), pain intensity (P value not reported).
Lopez-Rodriguez et al, ⁴⁶ 2007	Randomized controlled within-subject repeated measures	52 (35M) Grade 2 Chronic lateral ankle sprains	TCJ Manipulation (caudal) HVLA thrust + posterior gliding manipulation TCJ -HVLA thrust 1min	Placebo	Immediate	Proprioception	Proprioception significantly improved; load support—bilateral posterior load ($P=.016$), anterior load ($P=.04$), posterior load ($P=.043$), posterior-anterior load ($P=.016$).

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Table 1 (continued)

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Marron-Gomez et al, ¹⁵ 2015	RCT	52 (31M) Chronic ankle sprains	Weight-bearing MWM TCJ (posterior talar glide, posteroanterior tibial glide) 1 set of 10 repetitions TCJ HVLA distraction thrust ×3	Placebo	Immediate, short-term (24 and 48h)	Weight-bearing DFROM	MWM—Weight-bearing DFROM significantly greater than placebo — $P<.05$ (immediately and short-term). HVLA—weight-bearing DFROM significantly greater than placebo — $P<.001$ (immediately) and $P=.001$ (short-term).
Pellow and Brantingham, ⁴⁷ 2001	RCT	30 (19M) Grade 1/2 subacute and chronic lateral ankle sprains	Ankle axial elongation (TCJ— superior inferior by a mortise separation) 8 manipulations over 4wk	Detuned ultrasound treatment	Short-term (1-mo follow-up)	Non—weight-bearing DFROM, pain threshold, pain quality and intensity, function (Functional Evaluation Scale)	Non—weight-bearing DFROM significantly improved across time ($P=.001$) and between groups ($P=.001$). Pain threshold significantly improved across time ($P=.002$) and no significant group differences ($P=.395$). McGill significantly improved across time ($P=.001$) and between groups ($P=.004$). Pain intensity significantly improved across time ($P=.002$) and between groups ($P=.004$). Functional Evaluation Scale significantly improved across time ($P=.001$) and between groups ($P<.001$).
Plante, ⁴⁸ 2012	RCT	20 (12M) Chronic ankle sprains	TCJ (anteroposterior) 10 oscillations	Healthy subjects	Immediate	Weight-bearing DFROM, static balance, function (dynamic functional tasks)	Weight-bearing DFROM significantly improved across time ($P<.0001$). Single-limb stance, center of pressure significantly improved ($P<.04$). Dynamic functional task (center of pressure medial-lateral during jump task significantly improved [$P<.001$]; center of pressure medial-lateral during squat significantly improved [$P<.022$]; center of pressure medial-lateral during stance task significantly improved [$P<.039$]).

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Table 1 (continued)

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Reid et al, ⁴⁹ 2007	Randomized crossover	23 (8M) Chronic lateral ankle	Weight-bearing MWM (posterior talar glide, posteroanterior tibial glide) 10 repetitions × 2	Sham	Immediate	Weight-bearing DFROM	Weight-bearing DFROM significantly improved ($P=.02$).
Someeh et al, ⁵⁰ 2015	Experimental study design—within subjects	32 (20M) Chronic ankle sprains	Mulligan ankle taping/fibular repositioning taping	Healthy subjects	Immediate	Dynamic balance (SEBT)	Dynamic balance significantly improved across time—SEBT overall reach ($P=.001$).
Someeh et al, ⁵¹ 2015	Experimental study design—within subjects	32 (20M) Chronic ankle sprains	Not explicitly stated Mulligan ankle taping Not explicitly stated	Healthy subjects	Immediate	Function (dynamic functional tasks), participants' perceptions of stability and confidence	Function significantly improved across time; single leg hopping ($P=.014$); figure-of-8 hopping ($P=.05$); side hopping ($P=.001$). Confidence in abovementioned functional tests significantly improved across time; consequently, $P=.023$, $.048$, and $.038$, respectively.
Vicenzino et al, ¹⁶ 2006	Randomized crossover	16 (8M) Chronic lateral ankle sprains	Non-weight-bearing MWM (anteroposterior talar glide for dorsiflexion), 4 glides of 10s 4 sets Weight-bearing MWM (posterior talar glide, posteroanterior tibial glide) 4 sets of 10 glides	No intervention	Immediate	Weight-bearing DFROM, talar stiffness	Weight-bearing DFROM significantly improved ($P=.017$). Talar glide significantly improved ($P<.001$).
Wells, ⁵² 2012	RCT	17 (7M) Chronic ankle sprains	Maitland mobilization (TCJ—anteroposterior grade IV) 3 repetitions, 60s	No intervention	Immediate	Weight-bearing DFROM, non-weight-bearing DFROM, dynamic balance, pain intensity, static balance, stiffness, function (self-reported function)	Weight-bearing DFROM not significant ($P=.95$). Non-weight-bearing DFROM not significant ($P=.1$). Dynamic balance not significant; SEBT composite ($P=.8$); anterior ($P=.07$); posteromedial ($P=.79$); posterolateral ($P=.73$). Pain not significant ($P=.06$). Stiffness not significant ($P=.59$). Stability not significant ($P=.40$). Function (visual analog scale) not significant ($P=.44$).

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Table 1 (continued)

Study	Design	Sample	Intervention and Dosage	Comparator	Measurement Time Points	Outcomes	Results
Yeo et al, ⁵³ 2011	Randomized, controlled, within-subject repeated measures	13 (10M) Grade 2 Subacute lateral sprain	Maitland mobilization (distal TFJ anteroposterior) 3 sets of 1min mobilization	Placebo No intervention	Immediate	Weight-bearing DFROM, pressure-pain threshold, pain intensity, function (Functional Evaluation Scale)	Weight-bearing DFROM significantly improved ($P<.0001$). Pressure-pain threshold significantly improved ($P<.0001$). Pain intensity not significant ($P=.369$). Functional Evaluation Scale not significant ($P=.475$).

NOTE. "Immediate," measured immediately after the intervention. "Short-term," measured up to 3 months after the intervention. "Long-term," measured at 3 or more months after the intervention. Abbreviations: ADL, activities of daily living; FAAM, Foot and Ankle Ability Measure; HVLA, high velocity, low amplitude; M, male; RCT, randomized controlled trial; TCJ, talocrural joint; TFJ, tibiofibular joint.

Assessment of methodological quality

The methodological quality of individual studies was assessed using the PEDro Scale for randomized controlled trials and the Quality Assessment Tool for Observational Cohort and Cross-sectional Studies.³⁰⁻³² Two independent reviewers assessed the methodological quality, and the level of agreement between reviewers was assessed using Cohen's kappa.

Assessment of quality of evidence

The overall quality of evidence was assessed at the stage of meta-analysis, using the Grading of Recommendations, Assessment, Development, and Evaluation approach.³³ The quality of the evidence was classified as either high, moderate, low, or very low.³⁴ Risk of bias, consistency of results, directness (eg, generalizability), and precision (eg, sufficient data) were considered in assessing the overall quality.³⁵

Data extraction and statistical analysis

Descriptive data were extracted using an extraction table (table 1). Authors were contacted if possible where there were difficulties extracting data from the published article. Where feasible, study data that were comparable in terms of participant characteristics, outcome measures, and follow-up periods were pooled, and a meta-analysis was performed.

For the meta-analysis, the standardized mean difference (SMD) was calculated for the outcomes where the means and SDs were provided pre- and postintervention. This conversion of the data to a common scale permitted comparison of studies that used different tools to measure the same outcome. This review followed the general practice of interpretation for small, medium, and large effect sizes (0.2, small effect; 0.5, medium effect; 0.8, large effect).^{54,55} The mean difference (MD) was calculated for studies using the same instrument for measurement. The results were reported in forest plots with 95% confidence intervals (CIs). The minimal clinically important difference (MCID) was used to interpret the clinical meaningfulness of the findings. Inconsistency was quantified by calculating I^2 and interpreted as follows: 30% to 59% may represent moderate heterogeneity, 60% to 89% substantial heterogeneity, and 90% to 100% considerable heterogeneity between studies. If I^2 was $>30\%$, a random-effects model was used to incorporate intertrial heterogeneity.³¹

In the instance of multiple comparison groups, the sham group was selected as the control condition. For the outcome of "static balance," studies with eyes-closed balance were selected to maintain the homogeneity of the analysis. Further, in studies with multiple time points, measurements taken at 2 to 3 weeks were selected for the meta-analysis (eg, if effects were measured at the time points of 2d, 3wk, and 2mo in a single study, data from measurements at 3wk were selected for the analysis). All statistical analyses were conducted using RevMan 5.3.^b

Results

Selection and characteristics of included studies

The database search identified 1521 studies after duplicate removal, and a further 9 studies were identified through citation tracking and hand searching of reference lists (fig 1). After the first

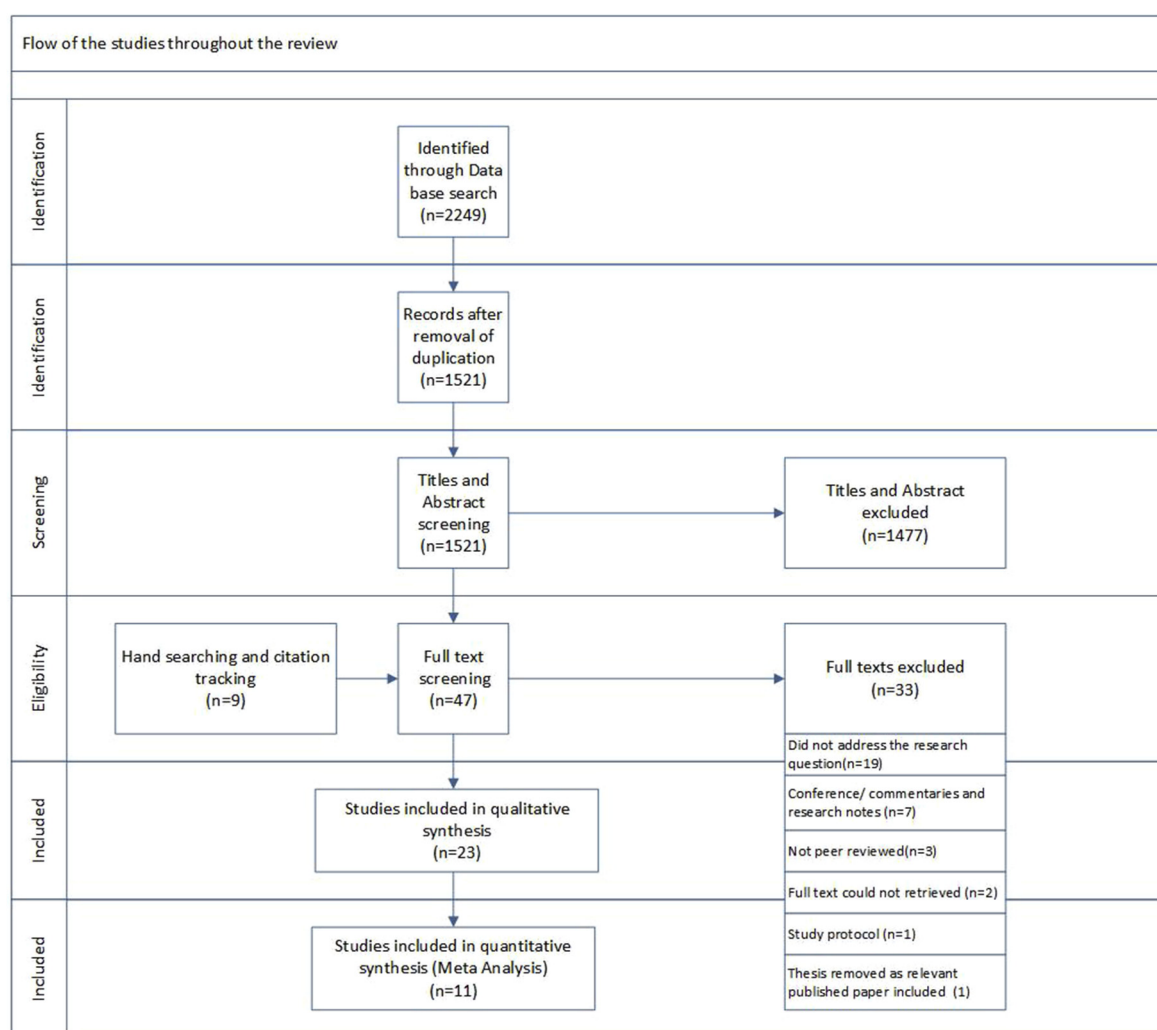


Fig 1 Flow chart of study selection.

stage of screening (using study title and abstract), 56 studies (database search, $n=47$; hand search, $n=9$) were identified as eligible for inclusion from the original 1530 (database search, $n=1521$; hand search, $n=9$) studies. Common reasons for exclusion after title and abstract screening included ineligible study design, joint mobilization not assessed in isolation, and study aim not relevant to the review research question. A further 33 studies were excluded in second stage (full text) screening, and reasons for exclusion included study aim not relevant to research question^{12,18,19,56-71} ($n=19$); conference proceedings, commentaries, and research notes⁷²⁻⁷⁸ ($n=7$); not peer reviewed⁷⁹⁻⁸¹ ($n=3$); full text not available^{82,83} ($n=2$); study protocol only⁸⁴ ($n=1$); and thesis removed as the relevant published article was included⁸⁵ ($n=1$). Twenty-three studies (including 3 theses) were therefore included in the current review. The interreviewer agreement for the title/abstract and full-text screenings was considered to be very good ($\kappa=.80$; 95% CI, .72–.89) and good ($\kappa=.71$; 95% CI, .52–.90), respectively. All disagreements were resolved by consensus. The data from 11 studies (including 2 theses^{48,52}) were available and deemed appropriate for inclusion in the meta-analysis (see [fig 1](#)). Publication bias was visually observed using funnel plots ([supplemental appendix S2](#), available online only at <http://www.archives-pmr.org/>).

The included studies were conducted in 7 countries (Australia, Canada, Iran, New Zealand, South Africa, Spain, United States) and involved a total of 585 participants. Twenty-one studies evaluated chronic ankle sprains, and 3 studies investigated sub-acute sprains. Outcomes measured varied widely and included dorsiflexion range of motion (DFROM), proprioception, stability/balance, pain threshold (pressure and thermal), pain intensity and quality, function, talar stiffness, postural sway, and patient confidence. A range of joint mobilization techniques were used, and these included MWM in weight bearing and/or non-weight bearing ($n=6$)^{13-16,21,49}; anteroposterior talocrural mobilization (Maitland grades III and IV)⁸⁶ ($n=4$)^{38,39,48,52}; high-velocity, low-amplitude ankle axial elongation manipulation and manipulation of the talocrural joint ($n=6$)^{15,36,44-47}; Mulligan ankle taping ($n=3$)^{42,50,51}; distal tibiofibular joint manipulation or mobilization ($n=2$)^{37,53}; and combined mobilization and traction of the talocrural joint ($n=4$)^{36,40,41,43}. Mulligan ankle taping was included because it aims to mimic a MWM by sustaining the fibula glide during daily activities.⁷ These techniques were variously applied by physiotherapists, medical practitioners, chiropractors, and athletic trainers. [Table 1](#) describes the participants, interventions, comparators, outcome measures, and results of the included studies.

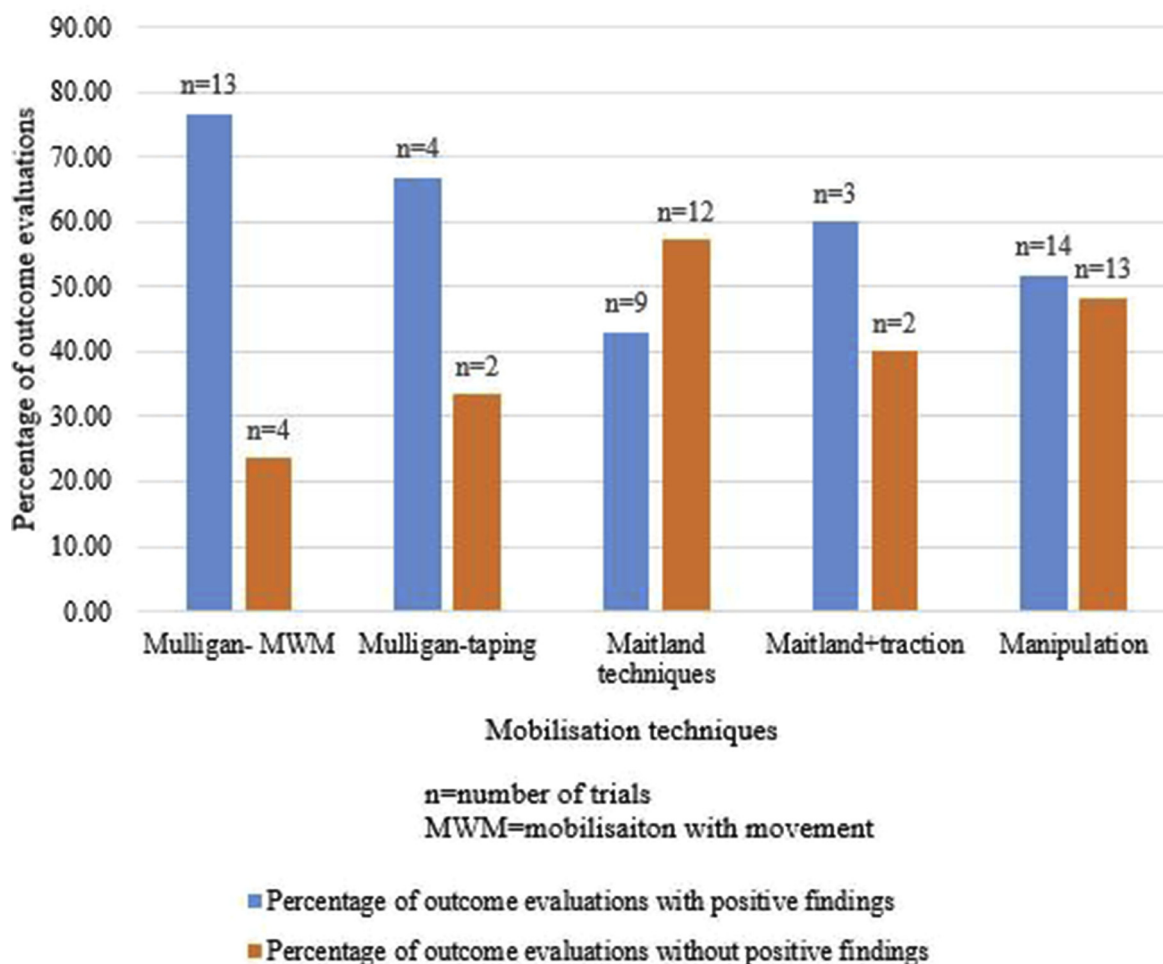


Fig 2 Percentage and number of outcome evaluations with and without positive findings following each technique combination of mobilisation for any clinically relevant outcome at any time point.

The immediate effects of joint mobilization were evaluated in 17 studies, short-term effects in 10 studies, and the long-term effects were assessed in only 1 study (see [table 1](#)). No studies evaluating effects on gait parameters, quality of life, reinjury rate, or strength were located in our search. In this systematic review, participants with chronic ankle sprains were included in 21 studies, and 3 studies included participants with subacute sprains. No studies measuring the effectiveness of mobilization in isolation for acute ankle sprains were able to be found. A meta-analysis was conducted using 11 studies, all involving participants with chronic ankle sprains.

Common mobilization techniques used in rehabilitation of ankle sprains

Five combinations of mobilization techniques were used in the 23 studies, including Mulligan MWM and taping techniques, Maitland mobilization with and without traction, and manipulation. The number of studies with positive effects on any clinically relevant outcome is contrasted against the number of studies with no positive effects, for each mobilization technique ([fig 2](#)). The findings also suggest that the combination of Mulligan MWM and taping is more likely to produce a clinical benefit than the other 3 mobilization combinations, as more (17)

of the studies using MWM techniques found positive outcomes compared with other techniques (Maitland mobilization 12, manipulation 14). Further, studies reporting no positive outcomes with MWM techniques are fewer in number (6) compared with the other techniques (Maitland mobilization 14, manipulation 13).

Quality of studies

Because of differences in study design, 2 tools were used to assess the methodological quality of the included studies. PEDro was used for the assessment of randomized controlled trials ($n=19$), and the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies was used for all other study types ($n=4$). The level of agreement between reviewers for the quality assessment was considered to be high ($\kappa=.63$; 95% CI, .53–.73), and all disagreements were resolved by consensus.

Most studies scored well on random allocation, adequate follow-up, and for providing both point measures and measures of variability for at least 1 key outcome. In studies assessed using the PEDro Scale ([fig 3](#)), the most common risk of bias was for therapist and subject blinding. For the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies, all 4 studies demonstrated bias in terms of insufficient time frame, different

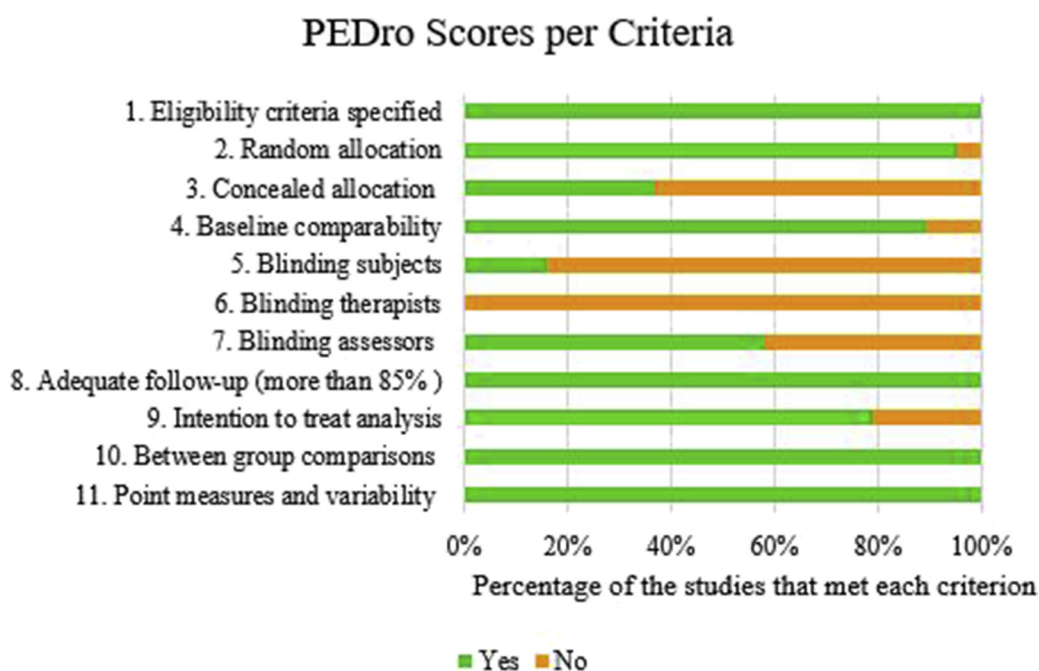


Fig 3 PEDro scores for assessment of quality of individual criteria.³⁰ Details about criteria: 1, Eligibility criteria were specified (Explanation: This criterion influences external validity, but not the internal or statistical validity of the trial. It has been included in the PEDro Scale so that all items of the Delphi Scale are represented on the PEDro Scale. This item is not used to calculate the PEDro score) (PEDro Scale); 2, Participants were randomly allocated to groups; 3, Allocation was concealed; 4, Groups were similar at baseline regarding most important prognostic indicators; 5, Blinding of all participants; 6, Blinding of therapists who administered the therapy; 7, Blinding of all assessors who measured at least 1 key outcome; 8, Measures of at least 1 key outcome were obtained from more than 85% of the participants; 9, All participants for whom outcome measures were available received the treatment or control condition as allocated; 10, Results of between-group statistical comparisons are reported for at least 1 key outcome; 11, Study provides both point measures and measures of variability for at least 1 key outcome.

levels of exposures as related to the outcome examined, and clearly defined valid and reliable exposure measures (fig 4). All studies scored at least moderate in terms of the overall quality of

the methodology for both the scales used (supplemental appendixes S3 and S4, available online only at <http://www.archives-pmr.org/>).

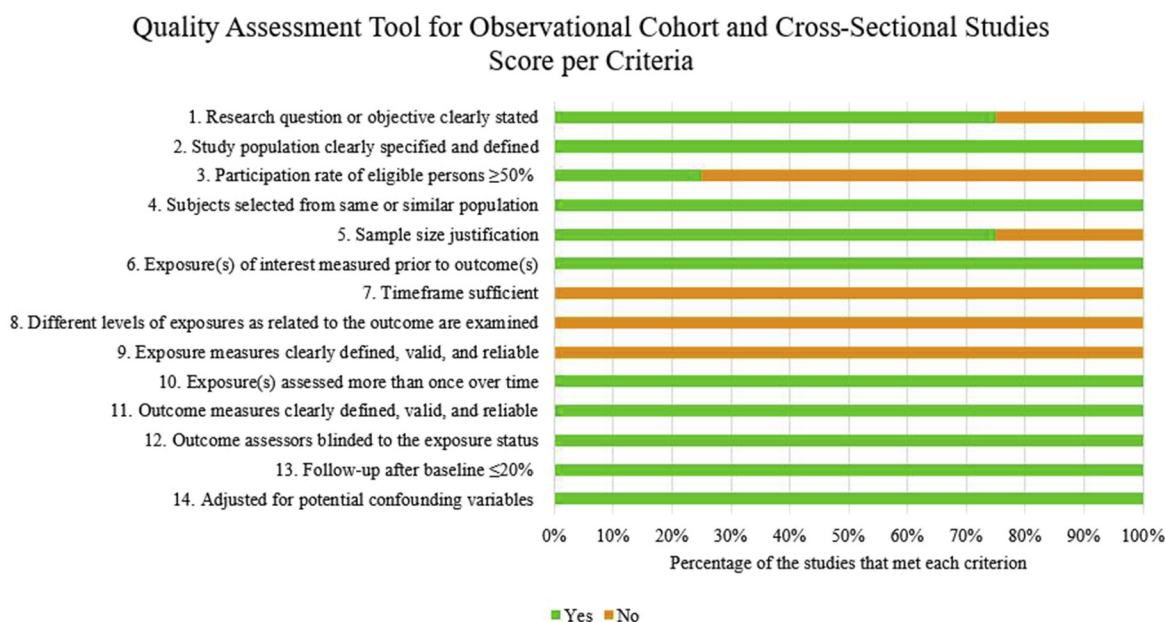


Fig 4 Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies.³²

Table 2 Number of outcome evaluations investigating at each time point of interest, listed by the reported positive effects

Outcome	Positive Findings					
	Immediate		Short-Term		Long-Term	
	Yes	No	Yes	No	Yes	No
1. DFROM	11	3	4	4	1	0
Weight-bearing DFROM	9	2	3	2	1	0
Non-weight-bearing DFROM	2	1	0	1	0	0
Unspecified	0	0	1	1	0	0
2. Proprioception	2	0	1	0	0	0
3. Stability/balance	3	7	3	3	1	0
Static balance	1	3	1	3	0	0
Dynamic balance	2	4	2	0	1	0
4. Pain threshold	1	1	1	1	0	0
5. Pain intensity	0	2	2	1	0	0
6. Functional outcomes	2	4	4	2	0	0
7. Talar stiffness	1	2	0	1	0	0
8. Recovery from postural sway	1	0	0	0	0	0
9. Patient's confidence toward stability	1	0	0	0	0	0

NOTE. "Immediate," measured immediately after the intervention. "Short-term," measured up to 3 months after the intervention. "Long-term," measured at 3 or more months after the intervention.

Effects of mobilization on subacute/chronic ankle sprains

The outcome measures of DFROM, proprioception, stability/balance, pain threshold, pain intensity and quality, function, talar stiffness, postural sway, and patient's confidence toward stability were assessed at varying time points across the studies after application of joint mobilization. Table 2 lists each outcome

evaluation, indicating positive effects of mobilization at each of the 3 time points of interest.

Eleven studies on chronic sprains reported quantitative data on 5 different outcomes, including weight-bearing DFROM, static balance, dynamic balance, pain intensity, and pain threshold. However, because of study heterogeneity and a lack of useable data for some outcomes, data could only be pooled for weight-bearing DFROM, static balance, dynamic balance, and pain

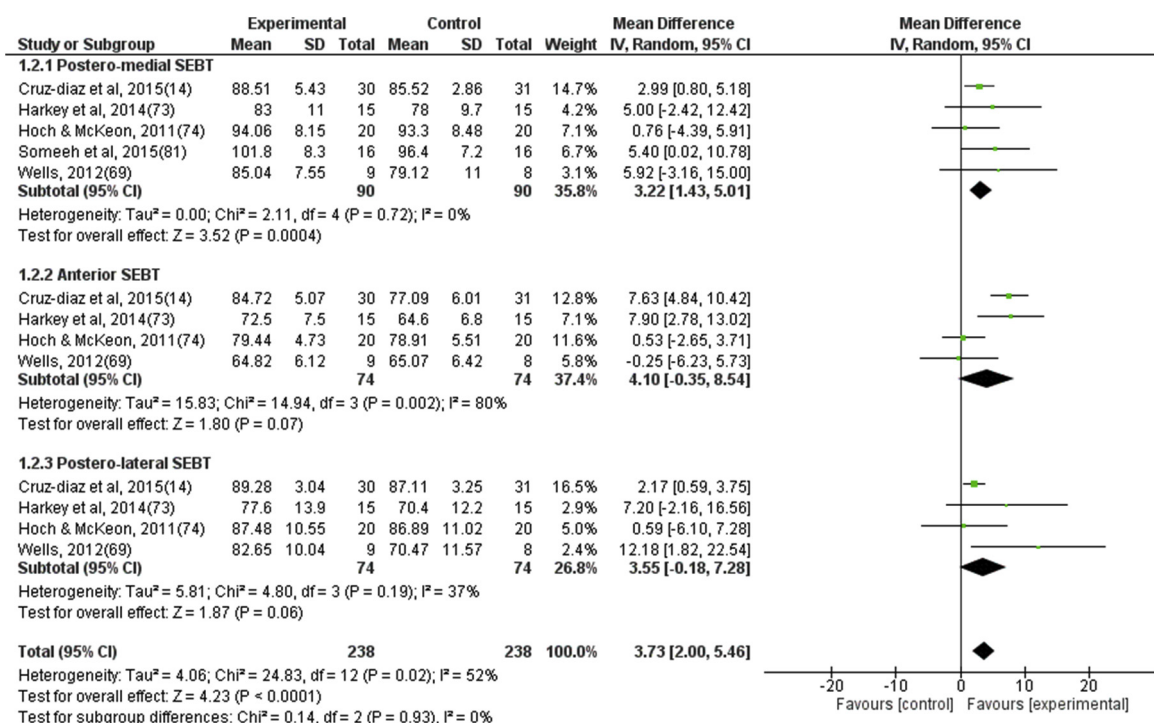


Fig 5 MD (95% CI) of the immediate effect of joint mobilization on dynamic balance by pooling data from 5 studies ($n = 180$). Abbreviations: df, degrees of freedom; IV, inverse variance.

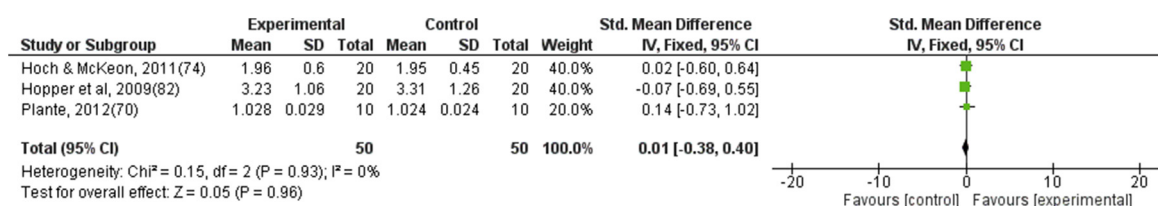


Fig 6 SMD (95% CI) of the immediate effect of joint mobilization on static balance by pooling data from 3 studies (n = 100). Abbreviations: df, degrees of freedom; IV, inverse variance; Std., standardized.

intensity in order to evaluate immediate effects, and weight-bearing DFROM was the only outcome measure available to assess the short-term effects of ankle mobilization.

Immediate effects of mobilization on ankle sprains

The immediate effects on DFROM were assessed in 14 outcome evaluations, of which 11 reported improvement with mobilization techniques (see table 2). The findings for other outcomes were less notable. Of the 10 studies that investigated the immediate effects of mobilization on stability/balance, 3^{14,39,50} had demonstrable improvement. Similarly, studies that assessed pain, talar stiffness, and function revealed inconsistent results. When considering the immediate effects of mobilization on functional outcomes, 2 outcome evaluations out of 6 demonstrated that it was effective.^{43,51} A summary of the reported immediate effects is provided in table 2.

Pooled data from 5 studies with a total of 180 participants were grouped for analysis of the effects of mobilization on each direction of the Star Excursion Balance Test (SEBT); anterior, posteromedial, and posterolateral. This analysis provided significant findings for the posteromedial direction of the SEBT (MD = 3.22; 95% CI, 1.43–5.01; *P* = .0004); however, the posterolateral direction (MD = 3.55; 95% CI, −.18 to 7.28; *P* = .06) and the anterior direction (MD = 4.10; 95% CI, −.35 to 8.54; *P* = .07) results of the SEBT were not significant (fig 5). Pooled data for static balance from 3 studies with a total of 100 participants indicated there were no significant immediate benefits after mobilization of individuals with chronic sprains when compared with control participants (SMD = .01; 95% CI, −.38 to .40; *P* = .96) (fig 6).

Similarly, data from 7 studies with a total of 249 participants indicated there were no significant immediate effects of mobilization on the weight-bearing DFROM of individuals with chronic sprains (SMD = .66; 95% CI, −.25 to 1.58; *P* = .16) (fig 7). For pain intensity, pooled data from 2 studies with a total of 47 participants indicated mobilization had no immediate effect on individuals with chronic sprains (SMD = −.21; 95% CI, −.78 to .37;

P = .48) (fig 8). There were insufficient data to analyze the immediate benefits of mobilization on pain threshold.

Short-term effects of mobilization on ankle sprains

Half of the outcome evaluations reported that mobilization improved DFROM, stability/balance, and pain threshold in the short-term (see table 2). Demonstrable improvement was also observed in pain intensity and function (see table 2), and 2 studies^{41,45} that evaluated short-term outcomes on talar stiffness and proprioception reported improvements. No studies reported short-term findings on postural sway or patient's balance confidence.

Pooled data from 2 studies with 94 participants with chronic sprains indicated joint mobilization was effective in the short-term for improving weight-bearing DFROM (MD = 2.56; 95% CI, .89–4.23; *P* = .003) (fig 9). There were insufficient data evaluating static balance, dynamic balance, pain threshold, and pain intensity to permit analysis of the short-term benefits of mobilization on these outcomes.

Long-term effects of mobilization on ankle sprains

Only 1 study¹⁴ evaluated the long-term effects of mobilization on ankle sprains. Long-term improvement in DFROM and stability/balance were reported in the single included study.

Quality of evidence

According to the Grading of Recommendations, Assessment, Development, and Evaluation (supplemental appendix S5, available online only at <http://www.archives-pmr.org/>), the evidence for DFROM (immediate and short-term), static balance, and dynamic balance can be considered to be of moderate quality. The evidence for pain was considered to be of low quality because of lack of generalizability of 1 of the included studies. Overall, the evidence included in this meta-analysis was considered to be of

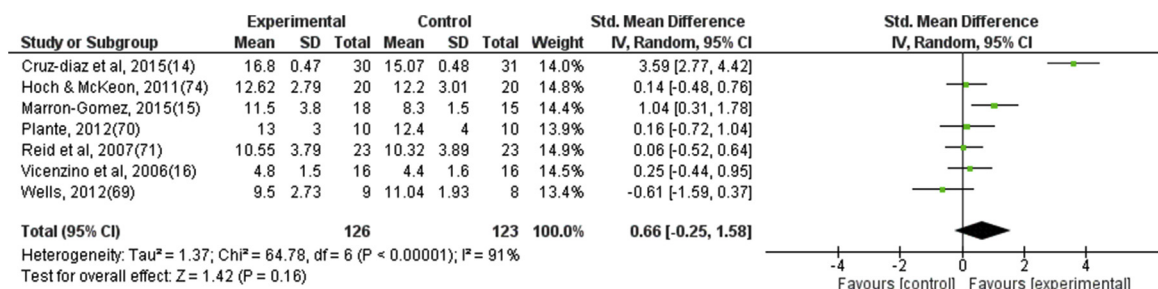


Fig 7 SMD (95% CI) of the immediate effect of joint mobilization on weight-bearing DFROM by pooling data from 7 studies (n = 249). Abbreviations: df, degrees of freedom; IV, inverse variance; Std., standardized.

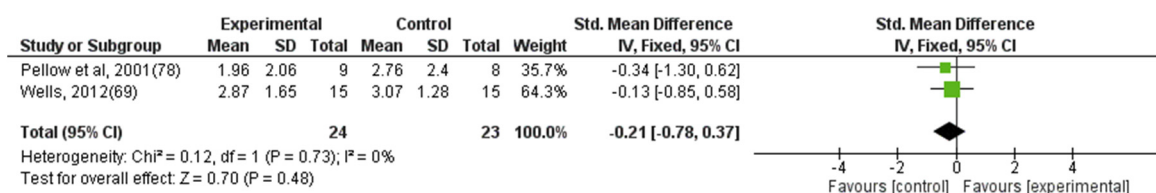


Fig 8 SMD (95% CI) of the immediate effect of joint mobilization on pain intensity by pooling data from 2 studies ($n = 47$). Abbreviations: df, degrees of freedom; IV, inverse variance; Std., standardized.

moderate quality, with the risk of bias and the level of heterogeneity the main factors influencing the quality of the evidence.

Discussion

This is the first systematic review to assess the clinical benefits of joint mobilization in the management of either lateral or medial ankle ligament sprains at all stages of recovery. Importantly, this is the first review to only include studies in which joint mobilization is the sole intervention. The current review did not identify any studies evaluating the clinical benefits of joint mobilization on acute ankle sprains, perhaps because mobilization is not typically the preferred choice of management in the acute stage of ankle sprains.⁸⁷ Findings about the clinical benefits of mobilization on most outcome measures were inconsistent across studies, and a lack of reported quantitative data, heterogeneity of subjects, and the differing types of joint mobilization applied made direct comparisons difficult. Despite this, meta-analysis indicated there are immediate benefits of mobilization for improving dynamic balance, and a short-term benefit in improving weight-bearing DFROM in chronic ankle sprains. These results provide compelling evidence that joint mobilization may be effective in improving balance immediately and in increasing DFROM in the short-term in chronic ankle sprains.

Dynamic balance and weight-bearing DFROM improvements after joint mobilization were both associated with clinically meaningful changes. The modified SEBT assesses performance during single-leg balance with reaching in 3 directions (anterior, posteromedial, posterolateral).^{88,89} The MCID for this test is reported as being 3.5%, and therefore the immediate effect on dynamic balance found in the meta-analysis ($MD = 3.73$) can be considered as clinically meaningful.^{88,89} It is plausible that the immediate improvements in dynamic balance after joint mobilization may increase the individual's balance confidence and perhaps reduce the risk of reinjury. Clinically, this may assist the individual with an ankle sprain to more safely proceed to the next level of functional exercise in the rehabilitation process.

There were no immediate improvements in either anterior SEBT performance or DFROM. Interestingly, previous research⁹⁰ supports the existence of a correlation between anterior SEBT

performance and the weight-bearing lunge test. This correlation could help explain the current review's findings on immediate anterior SEBT performance and DFROM. Notably, the MCID for ankle DFROM has not been established.⁹¹ However, approximately 3.6° of DFROM is associated with 1cm in distance from the wall in the lunge test.³⁹ The MD in the short-term measurement of weight-bearing DFROM from the current meta-analysis was 2.56cm, and this equates to 9.2° of dorsiflexion, which can be considered as clinically meaningful given that the normal total range is only 15° to 20° .^{92,93}

Joint mobilization techniques are aimed at restoring the normal joint ROM,^{94,95} and indeed this review found DFROM improved after mobilization. However, the mechanisms by which restoring ankle ROM may assist other impairments are unclear, as are the underlying mechanisms by which mobilization may actually work.^{15,16} It has been proposed that increased ankle ROM is due to the correction of a bony positional fault.¹⁰ It is further postulated that the correct alignment of the articular surfaces may help to restore normal biomechanics as well as sensorimotor function.¹⁰ However, it may be that mobilization produces less impact on pain, as evidenced by the lack of improvement in ankle pain outcome measures in this review. Potential underlying central nervous system changes related to persistent pain in chronic sprains remain unclear, but central sensitization may be a possible factor for persistence of chronic pain. If central sensitization is actually a key factor contributing to chronic ankle sprain pain, then changing the bony alignment would be unlikely to improve pain in chronic sprains since it is not the usual localized pressure-pain hypersensitivity⁹⁶ experienced immediately after a sprain.

According to the clinical practice guidelines linked to the *International Classification of Functioning, Disability and Health* from the American Physical Therapy Association, clinicians should use joint mobilization to improve ankle dorsiflexion, proprioception, and weight-bearing tolerance in patients recovering from a lateral sprain.⁸ Of these 3 outcomes, the findings of the current review only support the benefit of mobilization for dorsiflexion. There was insufficient research available to conclude whether mobilization is effective for improving proprioception or weight-bearing tolerance. However, the current review found clinically meaningful evidence

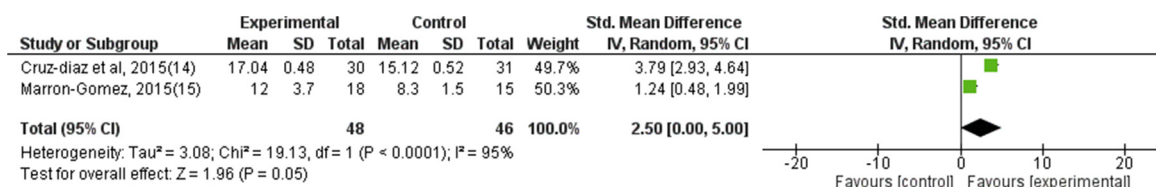


Fig 9 SMD (95% CI) of the short-term effect of joint mobilization on weight-bearing DFROM by pooling data from 2 studies ($n = 94$). Abbreviations: df, degrees of freedom; IV, inverse variance; Std., standardized.

for the effect of mobilization on dynamic balance, an outcome not mentioned in the clinical practice guidelines from the American Physical Therapy Association. One explanation for this difference may be that the guidelines only included literature published before April 2012, while the current review has included 7 more recently published studies.

The inclusion and exclusion criteria of the current review differ in important ways from previous systematic reviews on this topic. In contrast to those prior reviews, our search criteria included both lateral and medial ligament sprains, covered all stages of recovery from acute to chronic, and encompassed all clinically relevant outcomes used to assess the effects of mobilization. Importantly, of the 6 prior reviews^{22,24-27,97} that have evaluated the efficacy of mobilization techniques on ankle sprains, all included studies that did not evaluate joint mobilization as a unique intervention, but rather as an adjunct to other interventions (such as home exercise programs, RICE protocol, and external supports) included in their review. The current review excluded these multimodal studies to ensure the homogeneity of the included studies and to increase the precision of the results in relation to the effects of joint mobilization. Compared with the recent review by Loudon et al.²⁴ the present review included almost 3 times more studies (23), with all of these only investigating the clinical effects of joint mobilization techniques in isolation. In the review by Loudon,²⁴ only 8 studies were included, and of those, mobilization was used as the sole intervention in only 5.²⁴ This disparity in the number of included studies may be due to our searching a greater number of databases (11), including medial ankle sprains in the search criteria, by reviewing dissertations and theses, and by not limiting clinical outcomes.

This review includes the first meta-analysis undertaken to assess the clinical benefits of joint mobilization for ankle sprains. When comparing the findings of the current review to previous systematic reviews, there were some agreements and some inconsistent results. When considering the immediate effects of mobilization, the review by van der Wees et al.²⁶ reported an improvement in DFROM.²⁶ However, the current review did not support an immediate effect on weight-bearing DFROM, with mobilization providing only a short-term effect. Pain and function are concluded to improve immediately in the review by Southerst et al.,²⁷ but in our review immediate pain relief was not evident, and inconclusive results were found for immediate function. When considering the short-term effects, the effectiveness of mobilization in increasing ankle ROM was supported in the review of Bleakly et al.,²² and this was consistent with the findings of the current review.²² The review by van Ochten et al.²⁸ reported positive changes in short-term pain and function in chronic sprains; however, the findings of the present review were inconclusive for both of these outcomes. When considering the long-term effects of mobilization, pain and function are improved according to the review by Southerst.²⁷ The findings of the current review on these outcomes were inconclusive because of a lack of data. Different definitions of inclusion criteria for mobilization techniques included within reviews (eg, including other therapies such as home exercise or RICE treatment along with mobilization), as well as differences in the databases searched and the periods of the data searches, are all factors contributing to these differing findings.

Study limitations

Limitations of this review include the wide variation in follow-up time points that we defined as short-term (from 1d to <3mo).

Additionally, the included studies have used a range of different mobilization techniques and comparators. It was beyond the scope of this review to attempt to determine the independent merits of individual techniques. In particular, there may be value in analyzing joint mobilization and high-velocity thrust manipulation techniques separately rather than together, but given the lack of available research at this time directly comparing these 2 manual therapy approaches, this level of scrutiny is not possible. In addition, it was not possible to pool data to analyze the effectiveness of mobilization for some important outcomes that were reported in single studies. Despite attempts to contact the authors of included studies, data were insufficient to analyze the immediate effects on the pressure-pain threshold, and the short-term effects on the pressure-pain threshold and pain intensity. Finally, no high-quality evidence was found to provide robust evidence for the effectiveness of joint mobilization for ankle sprains.

Further research is required to determine the mechanisms by which mobilization improves dynamic balance and weight-bearing DFROM. Also, the long-term effects of mobilization on ankle sprains should be further investigated using clinically relevant outcomes.

Conclusions

Joint mobilization appears to clinically benefit individuals with chronic ankle sprains, improving dynamic balance immediately and weight-bearing DFROM in the short-term. It is unlikely to have an immediate effect on static balance, pain intensity, and weight-bearing DFROM. Other clinical outcomes that have been reported after mobilization demonstrate an inconsistent response to mobilization, and this may be a reflection of previous study designs or of the intervention itself.

Suppliers

- EndNote X7.3.1; Clarivate Analytics.
- RevMan (Review Manager) 5.3; The Nordic Cochrane Centre, The Cochrane Collaboration, 2014. Available at: <http://ims.cochrane.org/revman>.

Keywords

Ankle injuries; Ankle joint; Joint instability; Musculoskeletal manipulations; Physical therapy modalities; Rehabilitation

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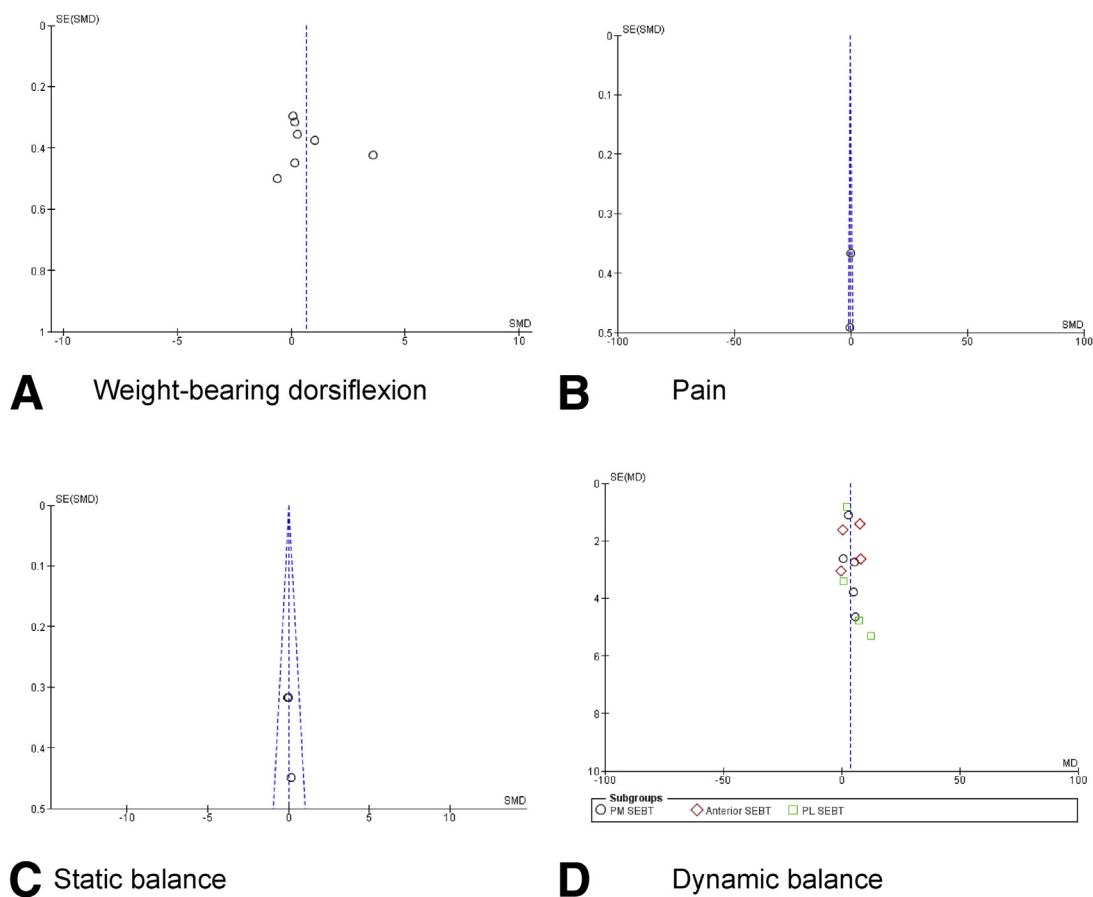
Supplemental Appendix S1 Search strategy

Searches

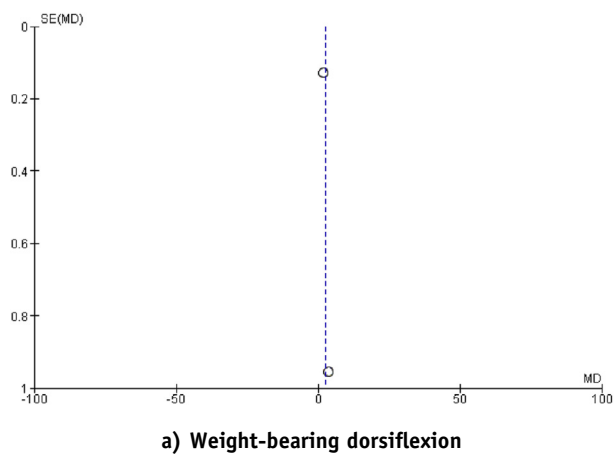
- 1 Ankle Injuries/
- 2 ankle sprain.mp.
- 3 (ankle* adj5 injur*).tw.
- 4 (ankle* adj5 sprain*).tw.
- 5 (ankle* adj5 twist*).tw.
- 6 (injur* adj5 ligament*).tw.
- 7 lateral ligament*.mp. or Collateral Ligaments/
- 8 Ankle Joint/or medial ligament*.mp.
- 9 Ankle Joint/or deltoid ligament*.mp.
- 10 ATFL.mp.
- 11 PTFL.mp.
- 12 "Sprains and Strains"/
- 13 talo crural.tw.
- 14 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or
10 or 11 or 12 or 13
- 15 Chiropractic/or Manipulation, Orthopedic/
- 16 musculoskeletal manipulation.mp. or Musculoskeletal
Manipulations/
- 17 (joint* adj5 manipul*).tw.
- 18 (ankle* adj5 rehab*).tw.
- 19 Mulligan*.mp.
- 20 Maitland*.mp.
- 21 MWM*.mp.
- 22 manual therap*.mp.
- 23 manual technique*.mp.
- 24 (joint* adj5 mobili?ation*).tw.
- 25 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24
- 26 Randomized controlled trial.pt.
- 27 clinical trial.pt.
- 28 random*.tw.
- 29 trial*.tw.
- 30 group*.tw.
- 31 case series.tw.
- 32 cross-over studies/
- 33 Cross-Sectional Studies/
- 34 exp Cohort Studies/
- 35 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34
- 36 14 and 25 and 35
- 37 limit 36 to humans

Supplemental Appendix S2 Funnel Plots

Immediate effect of mobilization on weight-bearing dorsiflexion, pain, static balance, and dynamic balance



Short-term effect of mobilization on weight-bearing dorsiflexion



Abbreviations: MD, mean difference; PL, posterolateral; PM, posteromedial; SEBT, Star Excursion Balance Test; SMD, standardized mean difference.

Supplemental Appendix S3 PEDro scores for assessment of quality of individual intervention studies

Study	PEDro Scale											Total Score Out of 10
	1) Eligibility Criteria	2) Random Allocation	3) Concealed Allocation	4) Baseline Comparability	5) Blinding Subjects	6) Blinding Therapists	7) Blinding Assessors	8) Adequate Follow-up (>85% of Subjects)	9) Intention- to-Treat Analysis	10) Between- Group Comparisons	11) Point Measures and Variability	
Alanson, ³⁶ 2012	+	+	+	—	—	—	+	+	+	+	+	7
Beazell et al, ³⁷ 2012	+	+	—	+	—	—	+	+	+	+	+	7
Collins et al, ¹³ 2004	+	+	—	+	+	—	+	+	—	+	+	7
Cruz-Diaz et al, ¹⁴ 2015	+	+	+	+	—	—	+	+	—	+	+	7
Harkey et al, ³⁸ 2014	+	+	+	+	—	—	+	+	+	+	+	8
Hoch and McKeon, ³⁹ 2011	+	+	+	+	—	—	+	+	+	+	+	8
Hopper et al, ⁴² 2009	+	+	—	+	—	—	—	+	+	+	+	6
Joseph et al, ⁴⁴ 2010	+	+	+	+	—	—	—	+	+	+	+	7
Kohne et al, ⁴⁵ 2007	+	+	+	—	—	—	—	+	+	+	+	6
Lopez-Rodriguez et al, ⁴⁶ 2007	+	—	—	+	—	—	—	+	+	+	+	5
Marron-Gomez et al, ¹⁵ 2015	+	+	—	+	+	—	+	+	+	+	+	8
Pellow and Brantingham, ⁴⁷ 2001	+	+	—	+	—	—	—	+	—	+	+	5
Plante, ⁴⁸ 2012	+	+	—	+	—	—	—	+	+	+	+	6
Reid et al, ⁴⁹ 2007	+	+	—	+	—	—	+	+	—	+	+	6
Someeh et al, ⁵⁰ 2015	+	+	—	+	—	—	—	+	+	+	+	6
Someeh et al, ⁵¹ 2015	+	+	—	+	—	—	—	+	+	+	+	6
Vicenzino et al, ¹⁶ 2006	+	+	—	+	+	—	+	+	+	+	+	8
Wells, ⁵² 2012	+	+	+	+	—	—	+	+	+	+	+	8
Yeo and Wright, ⁵³ 2011	+	+	—	+	—	—	+	+	+	+	+	7

NOTE. 1, Eligibility criteria were specified (Explanation: This criterion influences external validity, but not the internal or statistical validity of the trial. It has been included in the PEDro Scale so that all items of the Delphi Scale are represented on the PEDro Scale. This item is not used to calculate the PEDro score) (PEDro Scale); 2, Participants were randomly allocated to groups; 3, Allocation was concealed; 4, Groups were similar at baseline regarding most important prognostic indicators; 5, Blinding of all participants; 6, Blinding of therapists who administered the therapy; 7, Blinding of all assessors who measured at least 1 key outcome; 8, Measures of at least 1 key outcome were obtained from more than 85% of the participants; 9, All participants for whom outcome measures were available received the treatment or control condition as allocated; 10, Results of between-group statistical comparisons are reported for at least 1 key outcome; 11, Study provides both point measures and measures of variability for at least 1 key outcome.¹

Abbreviations: +, meet criteria; —, do not meet criteria.

Reference

1. PEDro Scale. 1999. Available at: <http://www.pedro.org.au/english/downloads/pedro-scale/>. Accessed May 12, 2015.

Supplemental Appendix S4 Quality assessment tool for observational cohort and cross-sectional studies scores for assessment of quality of individual cohort studies

Study	Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies Scale														Score Out of 14
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Gilbreath et al, ²¹ 2014	+	+	+	+	+	—	—	—	+	+	+	—	+	+	10
Hoch et al, ⁴⁰ 2012	+	+	+	+	+	—	—	—	+	+	+	+	+	+	11
Hoch et al, ⁴¹ 2014	+	+	+	+	+	—	—	—	+	+	+	—	+	+	10
Houston et al, ⁴³ 2013	+	+	+	+	—	—	—	—	+	—	+	—	+	—	7

NOTE. 1, Research question or objective clearly stated; 2, Study population clearly specified and defined; 3, Participation rate of eligible persons $\geq 50\%$; 4, Subjects selected from same or similar population; 5, Sample size justification; 6, Exposure(s) of interest measured before outcome(s); 7, Time frame sufficient; 8, Different levels of exposures as related to the outcome are examined; 9, Exposure measures clearly defined, valid, and reliable; 10, Exposure(s) assessed more than once over time; 11, Outcome measures clearly defined, valid, and reliable; 12, Outcome assessors blinded to the exposure status; 13, Follow-up after baseline $\leq 20\%$; 14, Adjusted for potential confounding variables. Total (0–14).^{1,2}
Abbreviations: +, meet criteria; —, do not meet criteria.

References

1. National Institutes of Health. Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies; 2014.
2. The Cochrane Collaboration. Cochrane handbook for systematic reviews of interventions; 2011.

Supplemental Appendix S5 Assessment of quality of evidence					
No. of Studies (Sample Size, n)	Risk of Bias	Inconsistency	Indirectness	Imprecision	Quality of Evidence
Immediate effects					
Outcome: DFROM					
7 studies (n; experimental=126: control=123)	Low risk of bias (PEDro scores: 6, 6, 7, 8, 8, 8, and 8)	<i>P</i> value on test for heterogeneity $P<.00001$, $I^2=91\%$ High inconsistency	Low indirectness	Low imprecision	Moderate quality (low risk of bias and high inconsistency)
Outcome: dynamic balance					
5 studies (n; experimental=90: control=90)	Low risk of bias (PEDro scores: 6, 7, 8, 8, and 8)	<i>P</i> value on test for heterogeneity $P=.02$, $I^2=52\%$ Moderate inconsistency	Low indirectness	Low imprecision	Moderate quality (low risk of bias and moderate inconsistency)
Outcome: static balance					
3 studies (n; experimental=50: control=50)	Moderate risk of bias (PEDro scores: 6, 6, and 8)	<i>P</i> value on test for heterogeneity $P=.93$, $I^2=0\%$ Low inconsistency	Low indirectness	Low imprecision	Moderate quality (moderate risk of bias and low inconsistency)
Outcome: pain intensity					
2 studies (n; experimental=24: control=23)	Moderate risk of bias (PEDro scores: 5 and 8)	<i>P</i> value on test for heterogeneity $P=.73$, $I^2=0\%$ Low inconsistency	Moderate indirectness (less generalizable)	Low imprecision	Low quality (moderate risk of bias, moderate inconsistency, and low indirectness)
Short-term effects					
Outcome: DFROM					
2 studies (n; experimental=48: control=46)	Low risk of bias (PEDro scores: 7 and 8)	<i>P</i> value on test for heterogeneity $P<.0001$, $I^2=95\%$ High inconsistency	Low indirectness	Low imprecision	Moderate quality (low risk of bias and high inconsistency)